# UNSUPERVISED EXPLAINABLE IMAGE ANOMALY DETECTION

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# EXECUTIVE SUMMARY

## Motivation

*Importance:* Critical for high-stakes industries (e.g., healthcare tumor identification, manufacturing quality control).

*Challenges:* Extreme data scarcity for anomalies; black-box models lack intrinsic explainability.

## Main Idea

- Engineered an unsupervised, patch-based feature extraction pipeline.
- Leveraged a pre-trained ResNet-18 to flag defects mathematically via spatial distance, completely eliminating the need for labeled anomaly data.

## Result

Achieved ROC-AUC: **0.9350** with native, pixel-precise explainability heatmaps.

# LITERATURE REVIEW

## RECONSTRUCTION & GENERATIVE MODELS (AUTOENCODERS, GANS)

*Mechanism:* Trained to reconstruct pristine images; assumes defects will cause a reconstruction failure.

*Limitation:* **Over-generalization.** Deep generative models are often too powerful. They successfully reconstruct the anomaly itself, reducing the error to zero and masking the defect.

## ONE-CLASS CLASSIFICATION (DEEP SVDD)

*Mechanism:* Maps all normal data points into a tightly bound mathematical sphere to establish classification boundaries.

*Limitation:* **Loss of Spatial Context.** Standard models pool the entire image into a single global vector. It successfully flags a defective image, but cannot natively localize the defect.

## FEATURE-BASED METHODS (THE CHOSEN ARCHITECTURE)

*Mechanism:* Maps localized image patches into a "nominal memory bank." Detects anomalies via direct Euclidean distance.

*Advantage:* **Intrinsic Explainability.** By calculating distance patch-by-patch, the spatial distance map inherently becomes the explanation, allowing for highly precise, native localization.

FEATURE EXTRACTION (VIA RESNET-18)

- Pass images through a frozen, pre-trained CNN.
- Extract intermediate layers to capture both local textures and global structures (inspired by PaDiM).

THE "NORMAL" MEMORY BANK

- Break the extracted features into localized spatial patches.
- Concatenate only the defect-free training patches to calculate the ultimate - absolute baseline of a 'perfect' product

DISTANCE-BASED INFERENCE

- Process test images through the same network.
- Calculate the exact Euclidean distance between each test patch and the normal

NATIVE EXPLAINABILITY

- Flag anomalies based on spatial distance thresholds.
- Map the patch distances directly into a high-resolution heatmap to visually isolate the defect.

# APPROACH

## THE FEATURE-MEMORY PIPELINE

# IMPLEMENTATION & DEMO

## CODE IMPLEMENTATION

https://github.com/Allison1212/i-hw-Spring-2026-mini-research/blob/main/Code_demo_final.ipynb

## SYSTEM DEMO

https://youtu.be/5FdHAq7infk

# REFERENCE


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